CARBON FOOTPRINT TRACKER FOR STUDENT LIFESTYLE USING AI

1M1B – IBM SkillsBuild AI for Sustainability Virtual Internship

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STUDENT DETAILS

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Internship: 1M1B – IBM SkillsBuild AI for Sustainability Virtual Internship

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ABSTRACT

Climate change is one of the most significant environmental challenges facing the world today. Human activities such as transportation, electricity consumption, food production, water usage, and consumer behavior contribute to increasing carbon emissions. Students are also contributors to carbon emissions through their daily lifestyle choices, yet many remain unaware of their environmental impact.

The Carbon Footprint Tracker for Student Lifestyle Using AI is an Artificial Intelligence-based sustainability solution developed to estimate, analyze, and reduce carbon emissions generated by students. The system uses Machine Learning techniques to analyze sustainability-related lifestyle data and predict carbon emissions. Based on the prediction results, the system provides sustainability recommendations that encourage environmentally responsible behavior.

This project supports Sustainable Development Goal (SDG) 13 – Climate Action and demonstrates the practical application of Artificial Intelligence for environmental sustainability.

INTRODUCTION

Background

Climate change is a global issue caused primarily by increasing greenhouse gas emissions. A carbon footprint represents the total amount of carbon dioxide and other greenhouse gases generated through human activities.

Students contribute to carbon emissions through:

- Transportation habits
- Electricity usage
- Food consumption
- Water consumption
- Housing conditions
- Spending behavior
- Recycling practices

Monitoring these activities can help individuals adopt sustainable lifestyle habits and reduce their environmental impact.

Need for the Project

Many students are unaware of how their daily activities affect the environment. Existing carbon footprint calculators often provide only basic calculations and lack personalized sustainability insights.

Challenges include:

- Limited awareness of carbon emissions
- Difficulty in estimating environmental impact
- Lack of personalized sustainability recommendations
- Limited use of Artificial Intelligence in sustainability education

An AI-powered system can provide accurate predictions and encourage environmentally responsible decision-making.

Objectives

The objectives of this project are:

- To estimate carbon emissions generated by students.
- To analyze sustainability-related lifestyle factors.
- To develop Machine Learning models for prediction.
- To provide sustainability recommendations.
- To promote climate action awareness.
- To support Sustainable Development Goals.

PROBLEM STATEMENT

Students generate carbon emissions through transportation choices, food consumption, electricity usage, water consumption, housing conditions, and spending behavior. However, most students do not understand the environmental impact of these activities.

Without proper awareness:

- Energy consumption increases.
- Unsustainable transportation practices continue.
- Carbon emissions remain high.
- Climate action goals become difficult to achieve.

Therefore, an Artificial Intelligence-based solution is needed to estimate student carbon footprints and encourage sustainable decision-making.

SDG ALIGNMENT

Primary SDG

SDG 13 – Climate Action

This project supports Climate Action by helping students understand, monitor, and reduce their carbon emissions through AI-powered prediction and sustainability insights.

Supporting SDGs

SDG 12 – Responsible Consumption and Production

Promotes responsible use of energy, water, transportation, and resources.

SDG 11 – Sustainable Cities and Communities

Encourages sustainable transportation and eco-friendly lifestyle choices.

SDG 4 – Quality Education

Creates awareness about environmental sustainability among students.

TARGET USERS

Primary Users

- College Students
- University Students

Secondary Users

- Educational Institutions
- Campus Sustainability Teams
- Environmental Organizations
- Researchers

WHY AI IS NEEDED

Traditional carbon footprint estimation methods are often manual and difficult to understand.

Artificial Intelligence helps by:

- Analyzing large sustainability datasets
- Identifying hidden patterns
- Predicting carbon emissions
- Generating sustainability insights
- Supporting data-driven environmental decisions

AI enables faster, scalable, and intelligent sustainability analysis.

AI SOLUTION OVERVIEW

The Carbon Footprint Tracker uses Artificial Intelligence and Machine Learning to estimate carbon emissions generated by students.

The system analyzes:

- Transportation Mode
- Food Habits
- Housing Type
- Electricity Usage
- Water Consumption
- Spending Behavior
- Recycling Habits

The Machine Learning model predicts the student's carbon footprint and identifies factors contributing to higher emissions.

The system then provides sustainability recommendations that help users reduce their environmental impact.

DATASET DESCRIPTION

Dataset Information

Dataset Name: Campus Sustainability and Carbon Emissions Data

Dataset Source: Kaggle

Dataset Link:
https://www.kaggle.com/datasets/zara2099/campus-sustainability-and-carbon-emissions-data

Dataset Size

- Total Records: 1000
- Total Features: 9

Target Variable

- Carbon_Emissions_kgCO2

Features Used

- Transportation_Mode
- Food_Habit
- Housing_Type
- Electricity_Usage
- Water_Consumption
- Spending_Behavior
- Recycling_Habit
- Sustainability_Score
- Carbon_Emissions_kgCO2

METHODOLOGY

Step 1: Data Collection

The dataset was collected from Kaggle and imported into the development environment.

Step 2: Data Cleaning

The dataset was cleaned by handling missing values, removing inconsistencies, and preparing the data for analysis.

Step 3: Data Preprocessing

Categorical variables were encoded and numerical variables were standardized where required.

Step 4: Exploratory Data Analysis

Various visualizations were generated to identify patterns, trends, and relationships between variables.

Step 5: Machine Learning Model Development

The following algorithms were used:

Linear Regression

Used as the baseline prediction model.

Decision Tree Regressor

Used to capture non-linear relationships within the dataset.

Random Forest Regressor

Used to improve prediction accuracy through ensemble learning.

Step 6: Model Evaluation

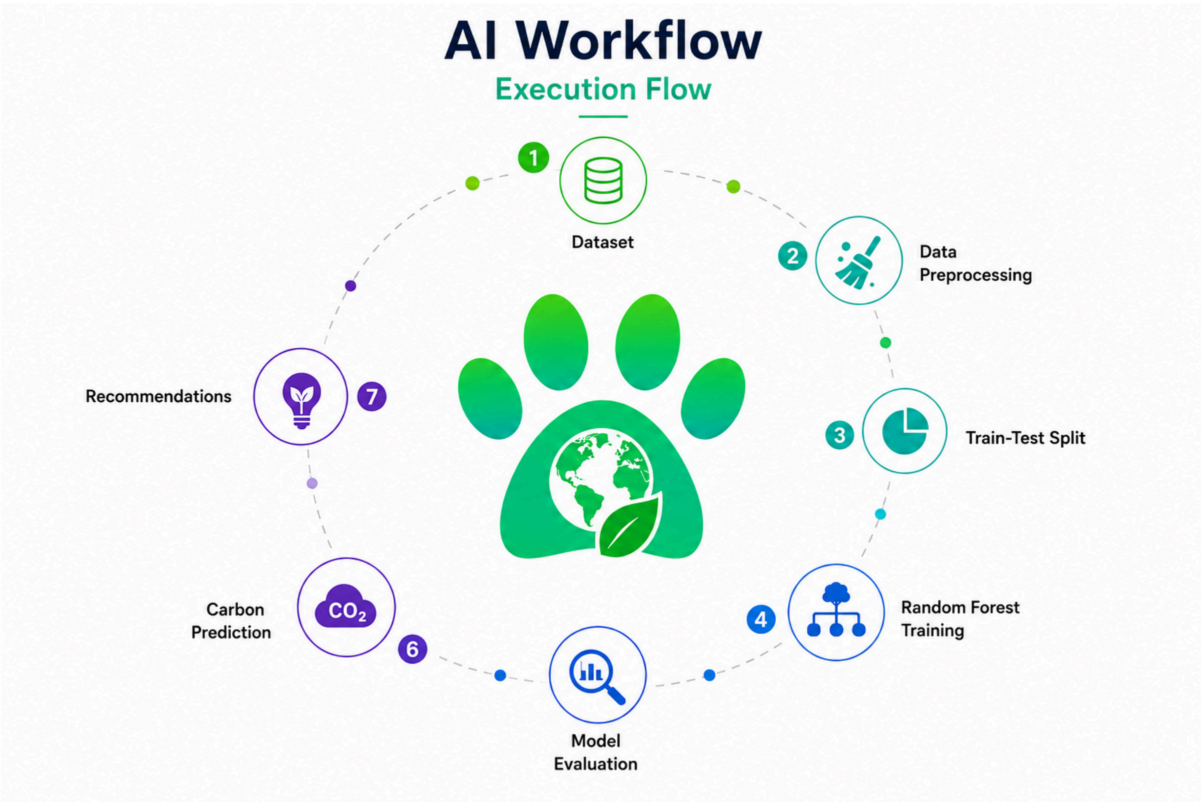
Models were evaluated using:

- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)

AI WORKFLOW

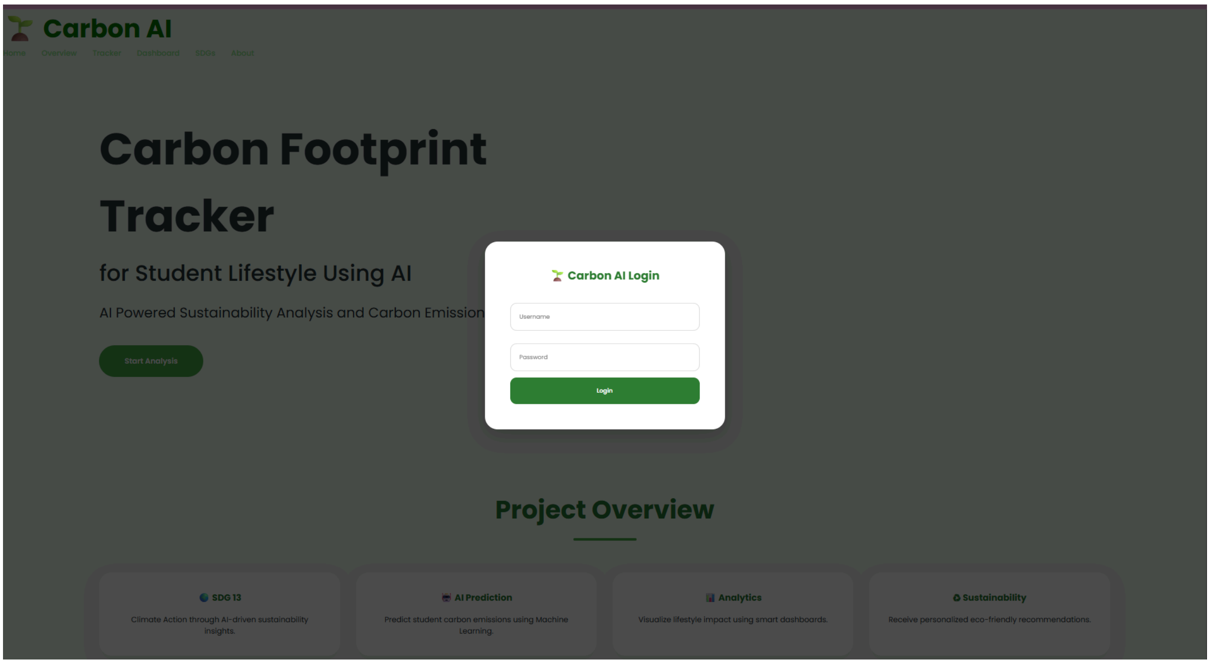
Student Lifestyle Data
↓
Data Collection
↓
Data Cleaning

↓
Feature Engineering
↓
Exploratory Data Analysis
↓
Machine Learning Model
↓
Carbon Emission Prediction
↓
Dashboard Visualization
↓
Sustainability Recommendations
↓
Climate Action Awareness



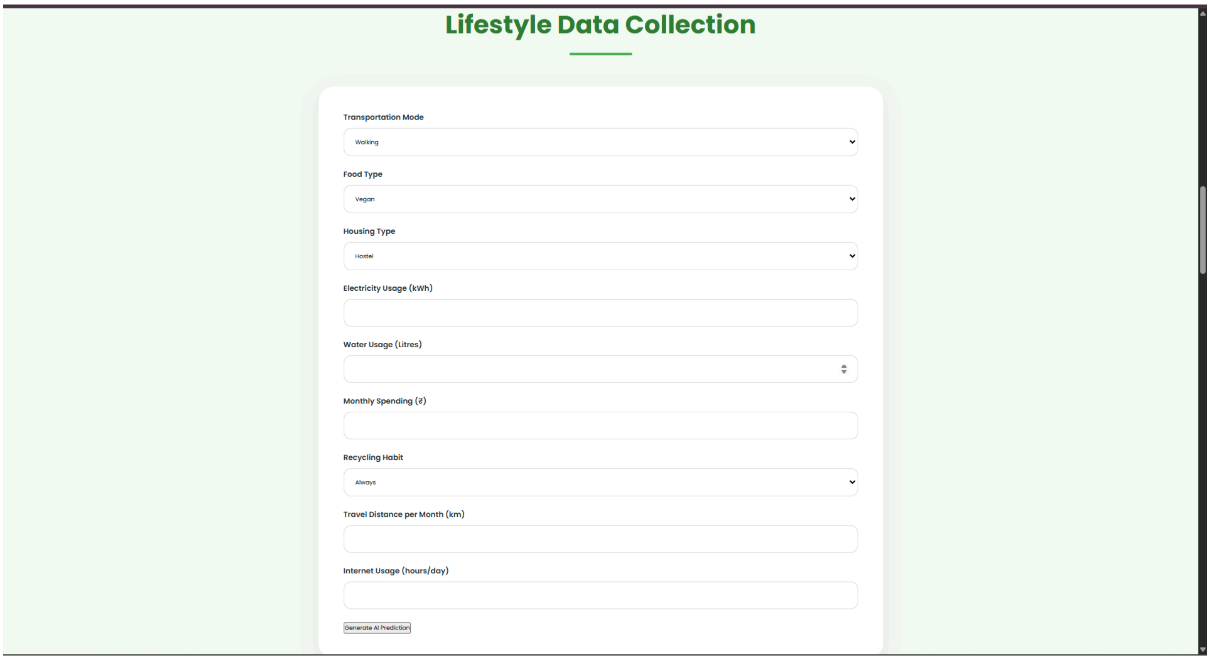
CHAPTER 10: PROJECT DEMONSTRATION

Home Page



The homepage introduces the project, sustainability objectives, and SDG alignment.

Student Input Form



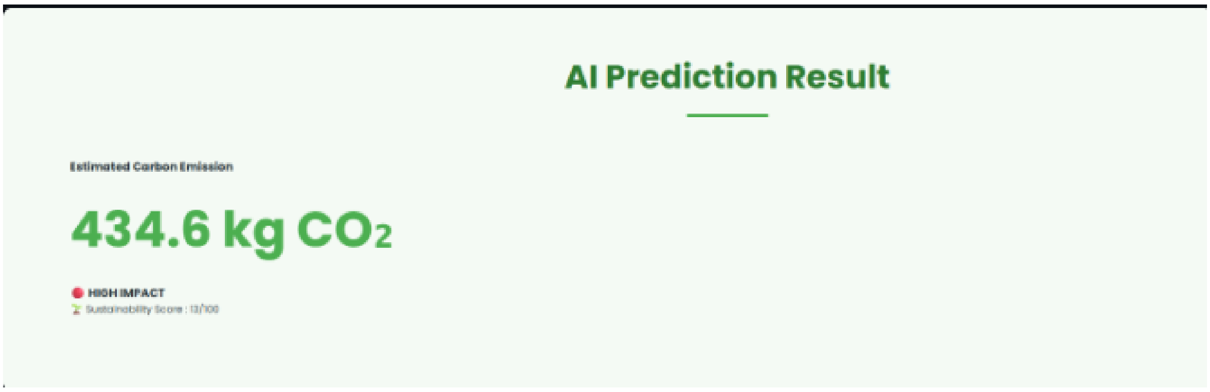
The input form collects sustainability-related lifestyle information.

Input Parameters

- Transportation Mode
- Food Habit

- Housing Type
- Electricity Usage
- Water Usage
- Spending Behavior
- Recycling Habit

Carbon Footprint Prediction Output

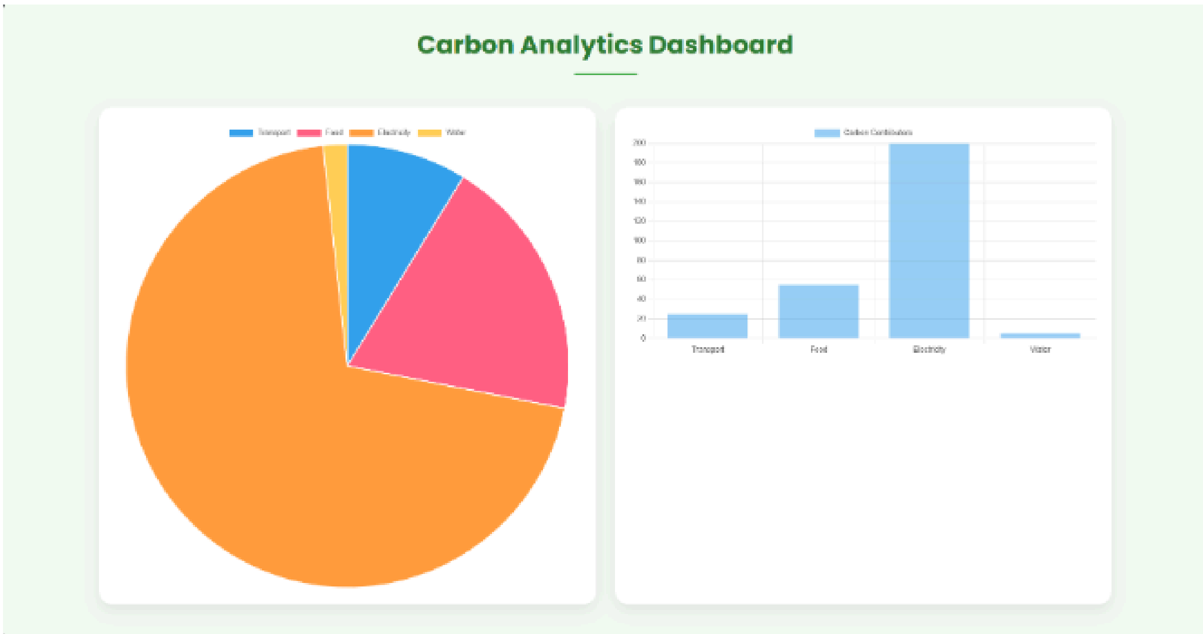


The Machine Learning model predicts the estimated carbon emissions generated by the student.

Output Includes

- Predicted Carbon Emission
- Sustainability Score
- Emission Category

Dashboard Analytics

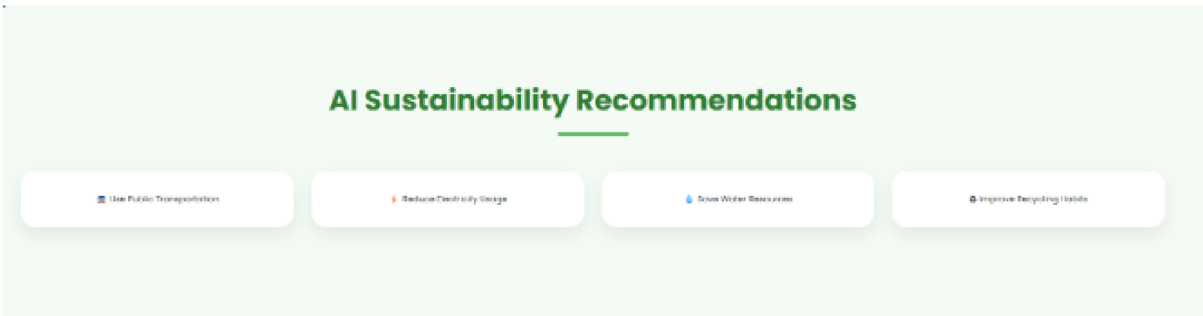


The dashboard provides visual sustainability insights.

Features

- Carbon Emission Analysis
- Transportation Analysis
- Food Habit Analysis
- Energy Consumption Analysis
- Sustainability Metrics

Sustainability Recommendations



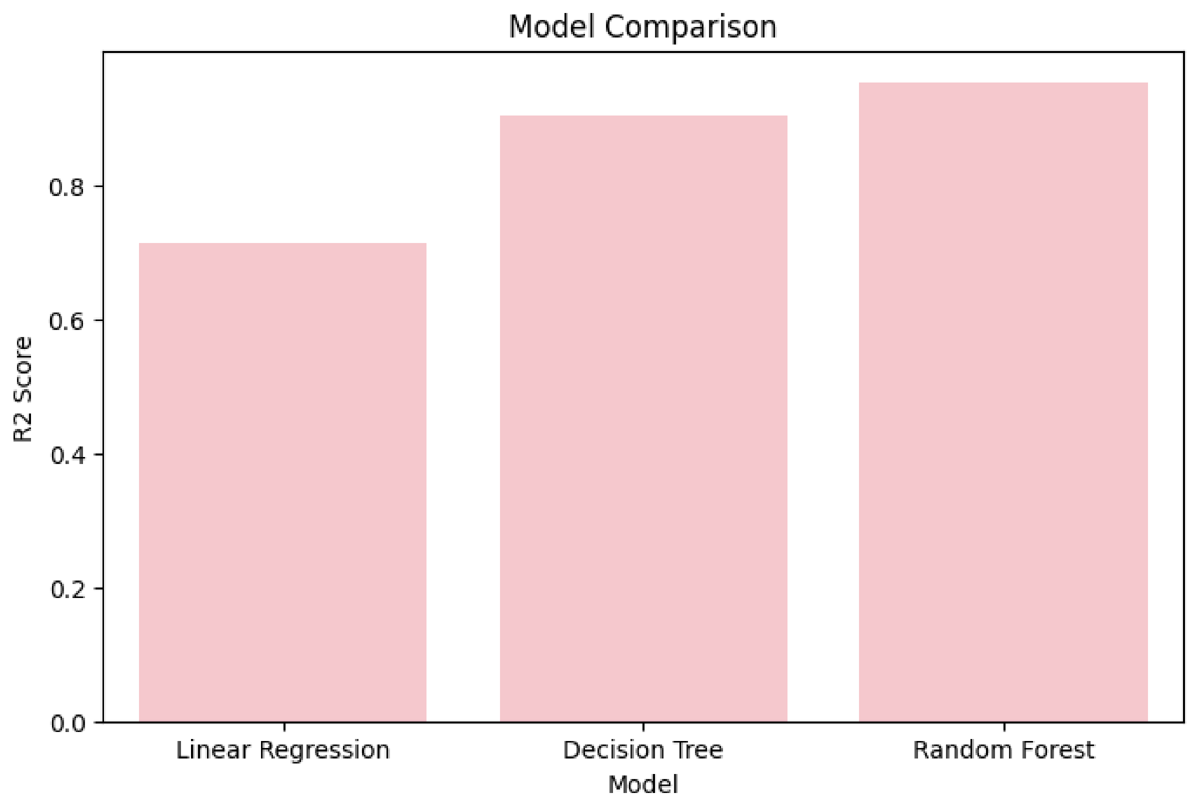
The system provides personalized sustainability recommendations.

Examples

- Use Public Transport
- Reduce Electricity Consumption
- Save Water
- Practice Recycling

- Adopt Sustainable Lifestyle Habits

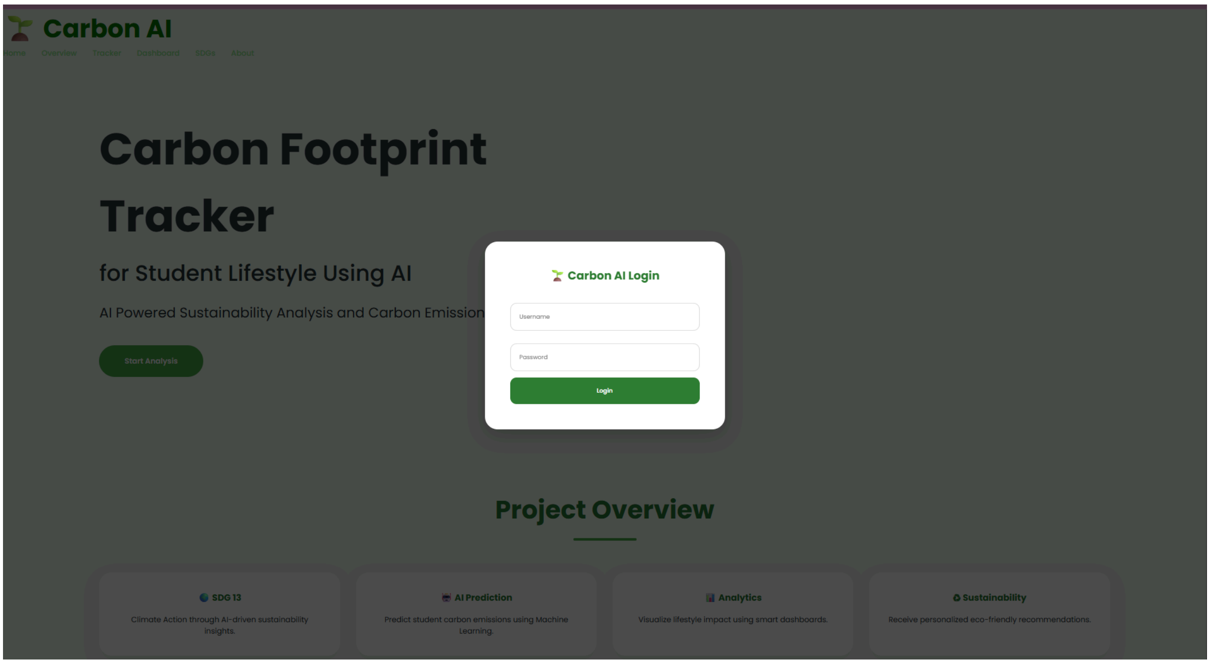
Machine Learning Model Performance

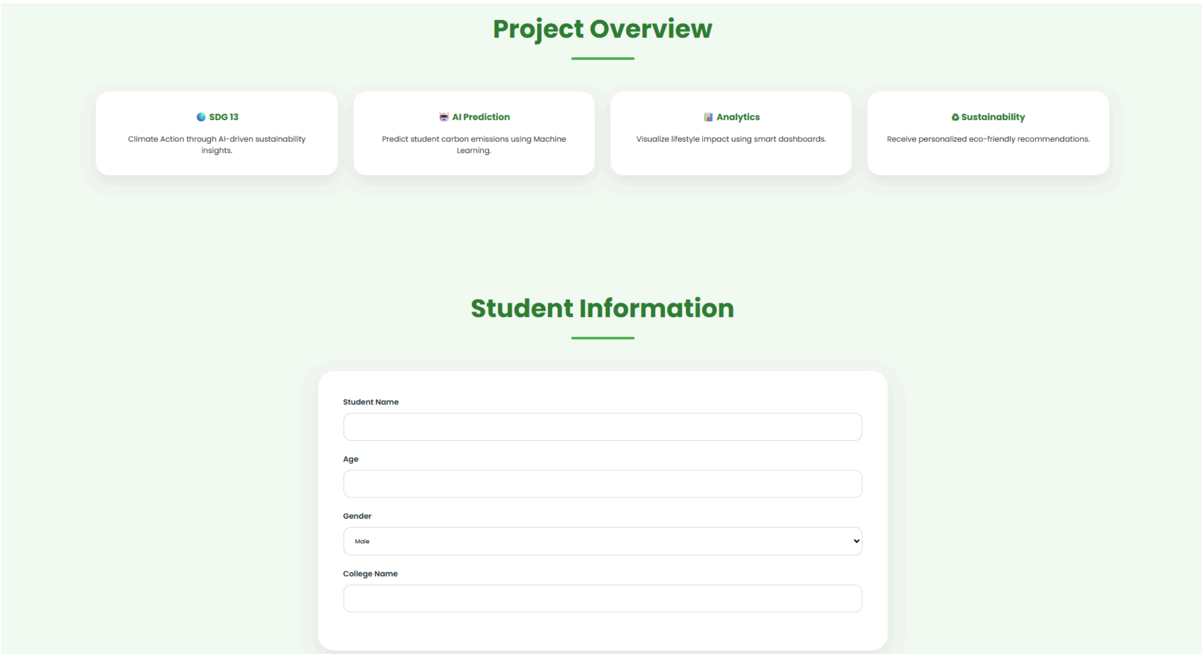


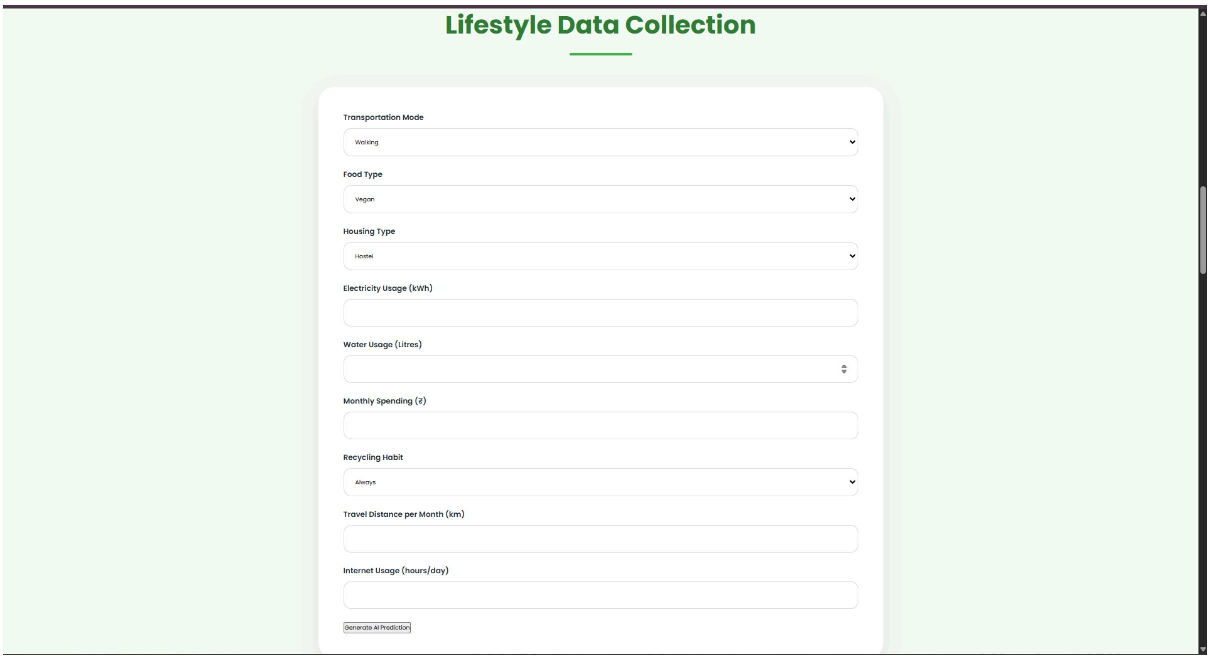
Performance Metrics

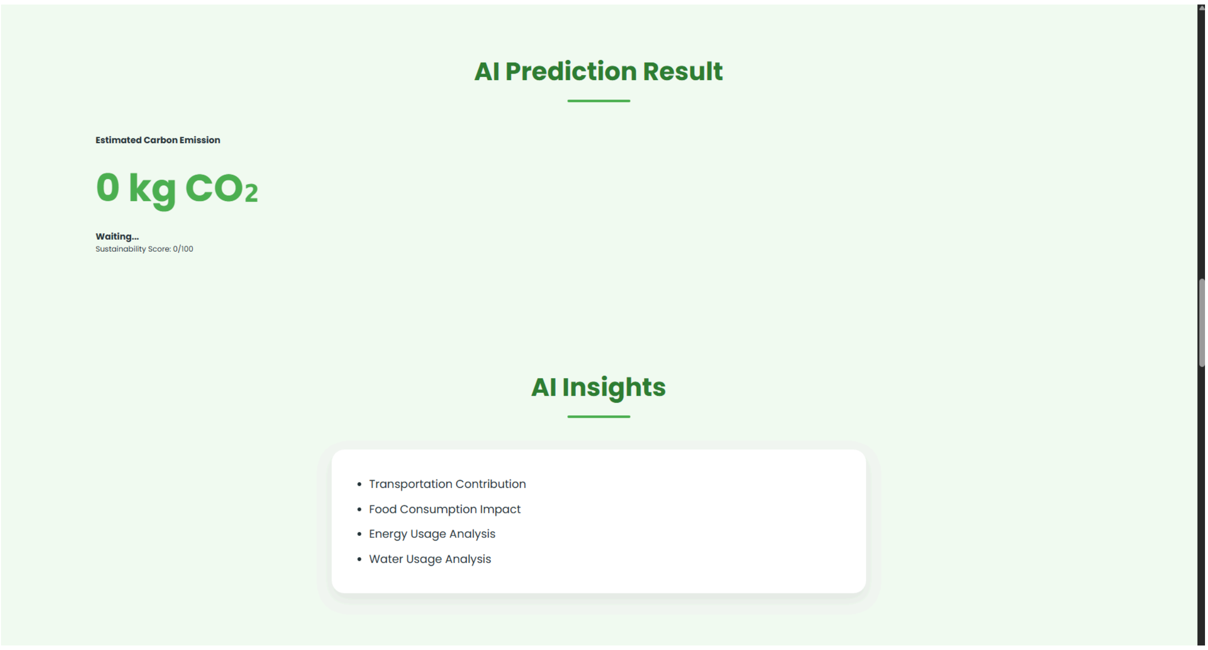
- R^2 Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Model Accuracy

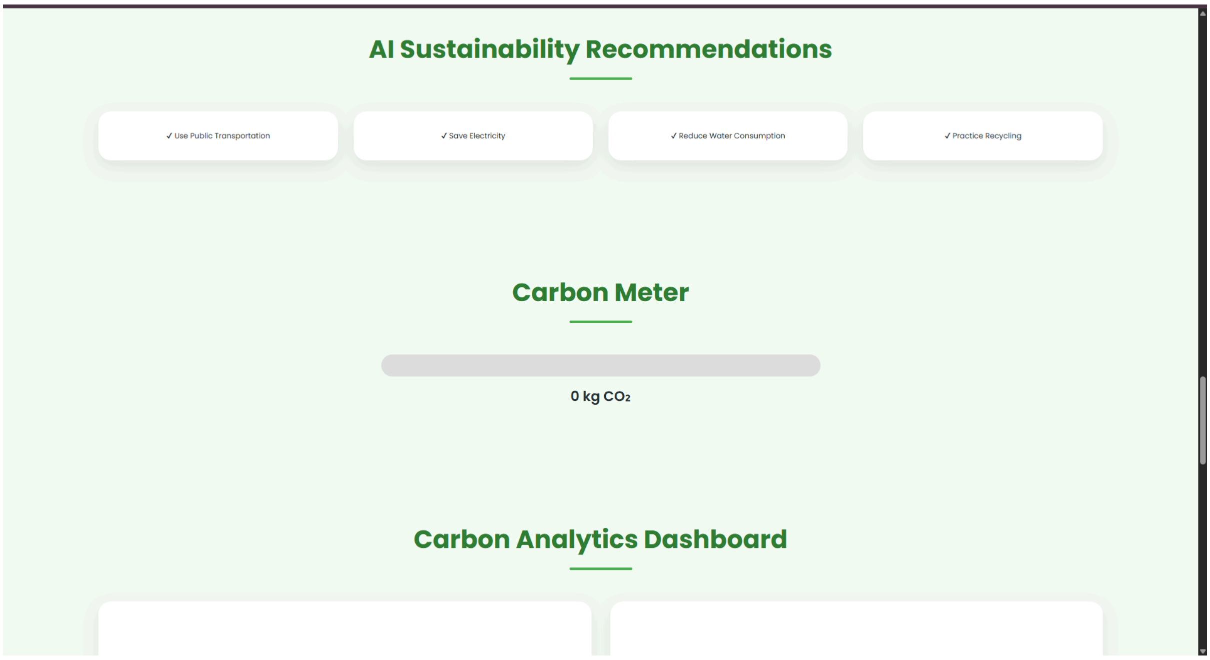
Complete Workflow

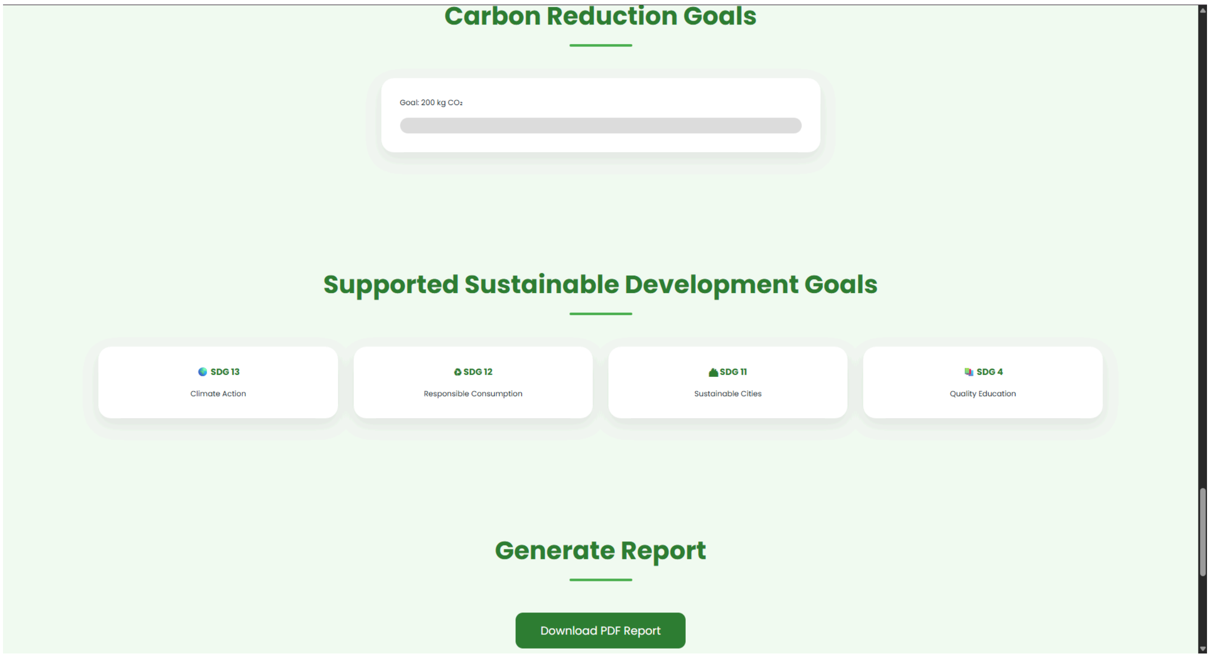

Carbon AI
Carbon Footprint Tracker
for Student Lifestyle Using AI
AI Powered Sustainability Analysis and Carbon Emission
Start Analysis
Carbon AI Login
Username
Password
Login
Project Overview
SDG 13
Climate Action through AI-driven sustainability insights.
AI Prediction
Predict student carbon emissions using Machine Learning.
Analytics
Visualize lifestyle impact using smart dashboards.
Sustainability
Receive personalized eco-friendly recommendations.


Project Overview
SDG 13
Climate Action through AI-driven sustainability insights.
AI Prediction
Predict student carbon emissions using Machine Learning.
Analytics
Visualize lifestyle impact using smart dashboards.
Sustainability
Receive personalized eco-friendly recommendations.
Student Information
Student Name
Age
Gender
Male
College Name


Lifestyle Data Collection
Transportation Mode
Walking
Food Type
Vegan
Housing Type
Hostel
Electricity Usage (kWh)
Water Usage (Litres)
Monthly Spending (₹)
Recycling Habit
Always
Travel Distance per Month (km)
Internet Usage (hours/day)
Generate AI Prediction


AI Prediction Result
Estimated Carbon Emission
0 kg CO2
Waiting...
Sustainability Score: 0/100
AI Insights
Transportation Contribution
Food Consumption Impact
Energy Usage Analysis
Water Usage Analysis

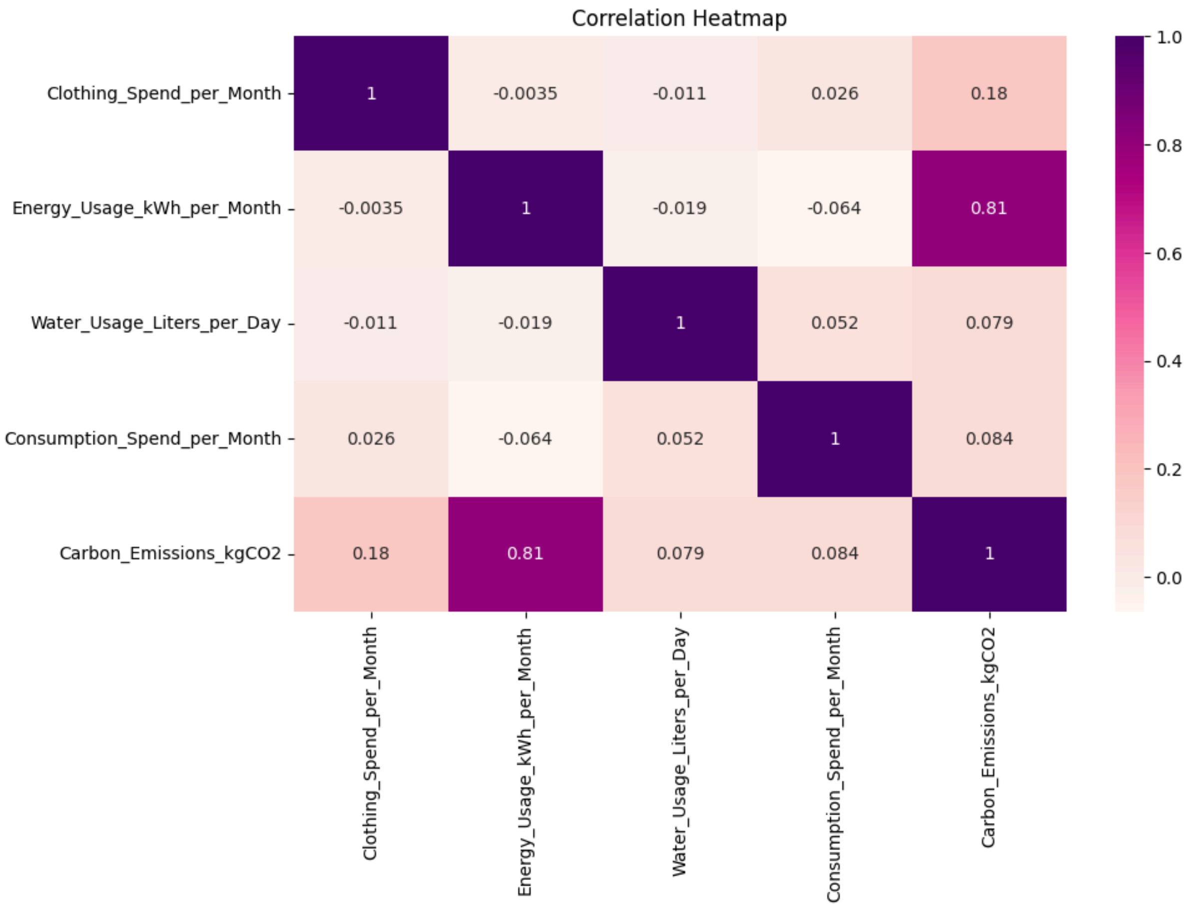




Displays the complete AI-based sustainability prediction workflow.

DATA VISUALIZATION AND ANALYSIS

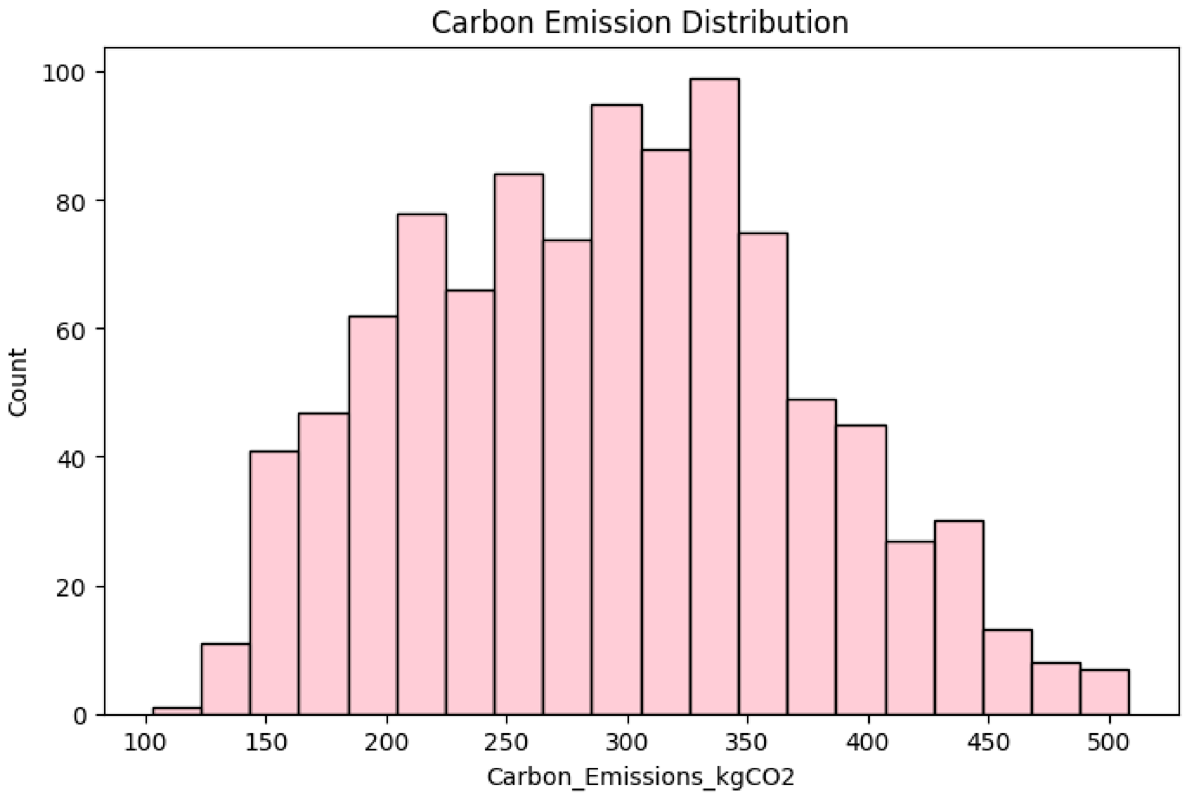
Correlation Heatmap



Purpose:

- Shows relationships between variables.
- Identifies factors affecting carbon emissions.

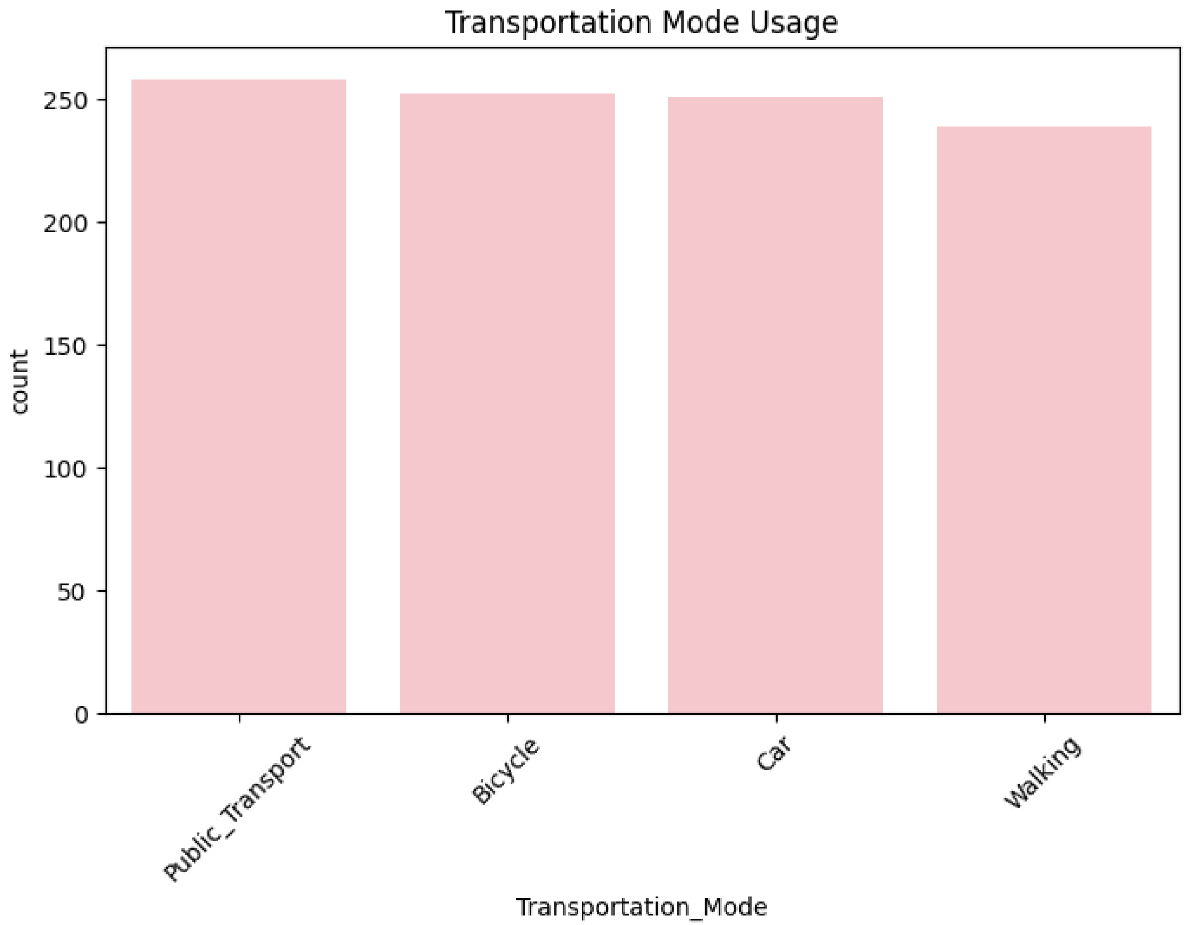
Carbon Emission Distribution



Purpose:

- Displays carbon emission distribution.
- Analyzes emission patterns.

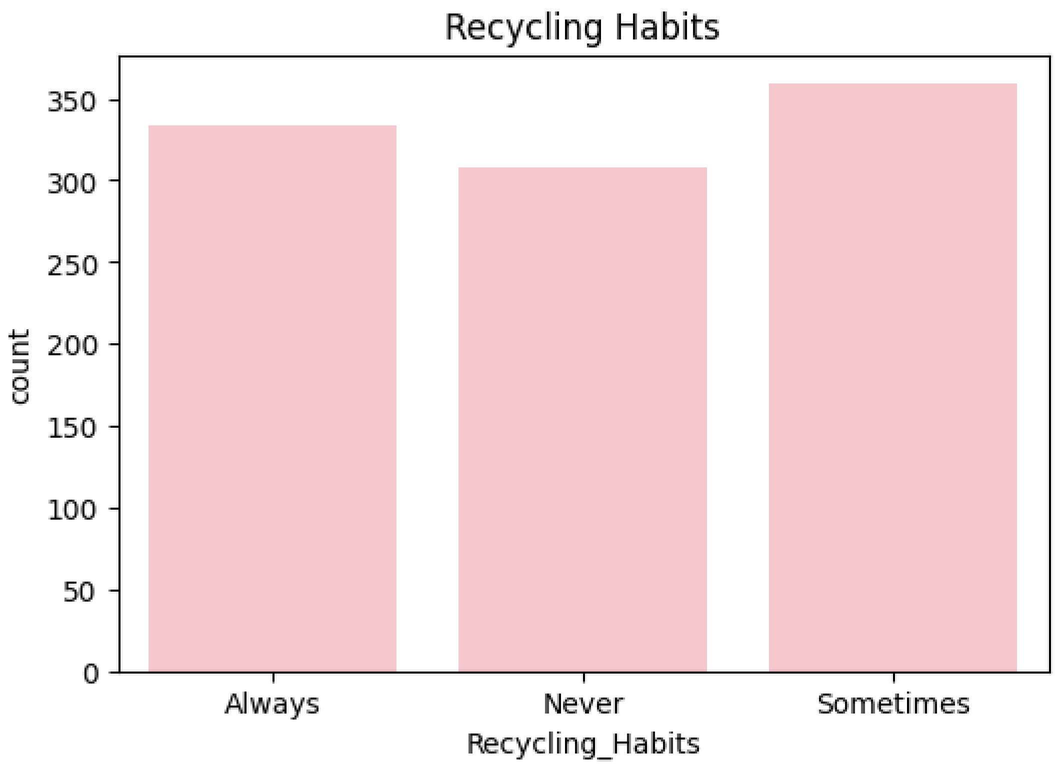
Transportation Mode Analysis



Purpose:

- Shows transportation trends.
- Highlights transportation impact.

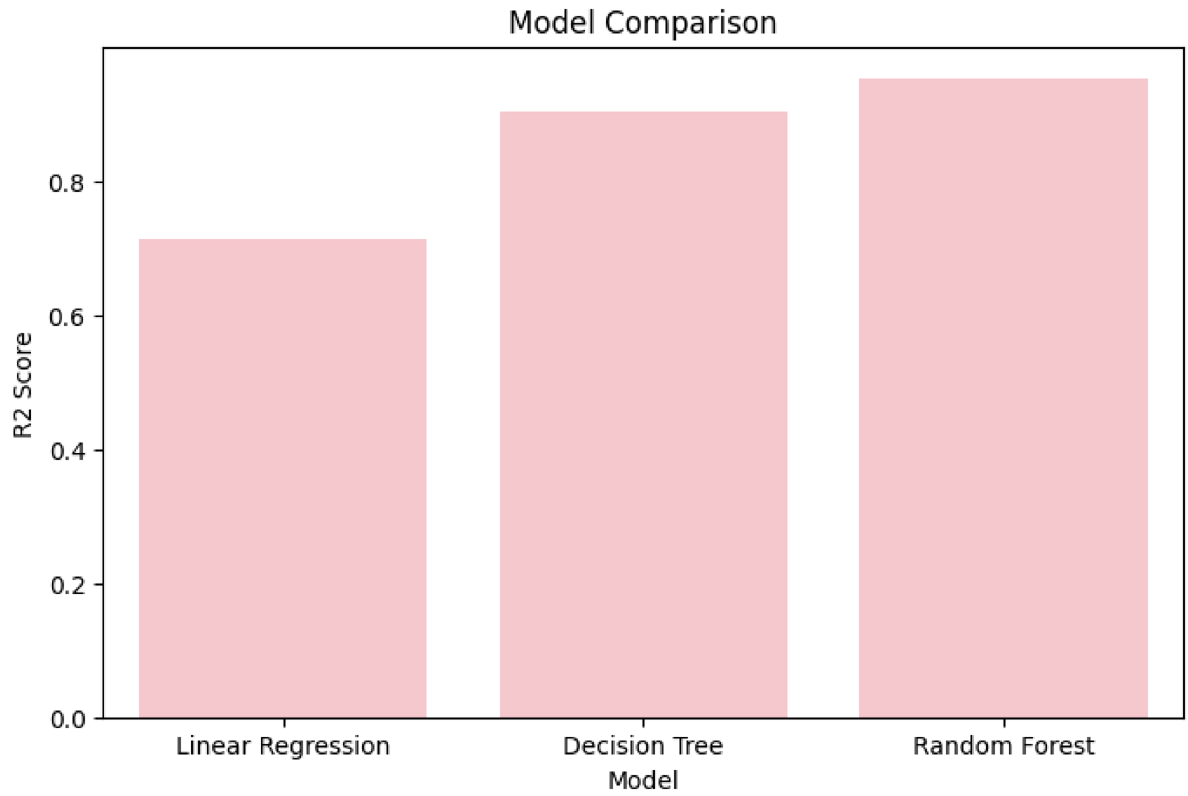
Recycling Habit Analysis



Purpose:

- Visualizes recycling behavior.
- Evaluates sustainability practices.

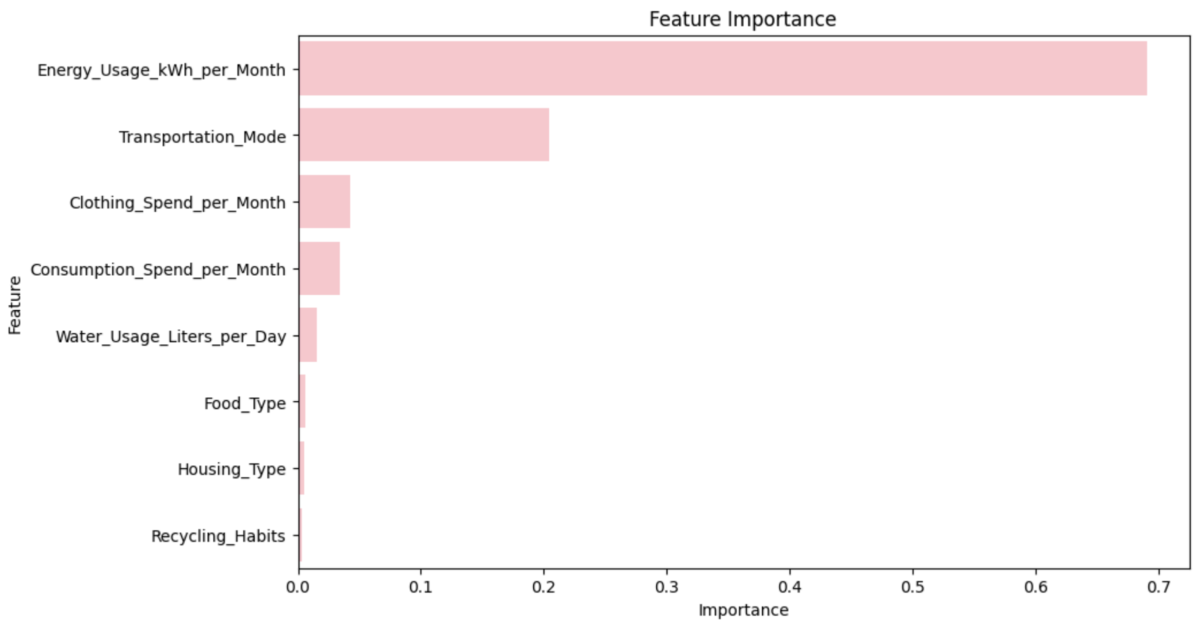
Machine Learning Model Comparison



Purpose:

- Compares model performances.
- Justifies model selection.

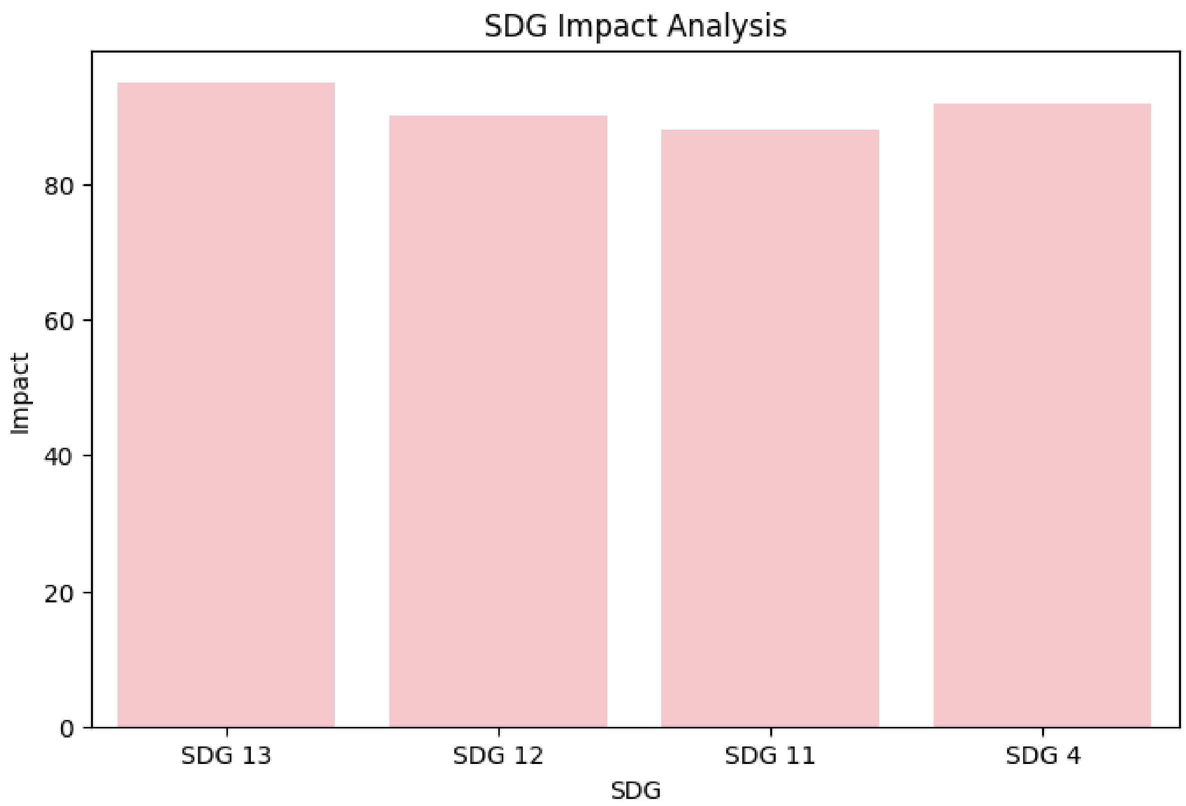
Feature Importance Analysis



Purpose:

- Identifies important prediction factors.
- Supports Explainable AI.

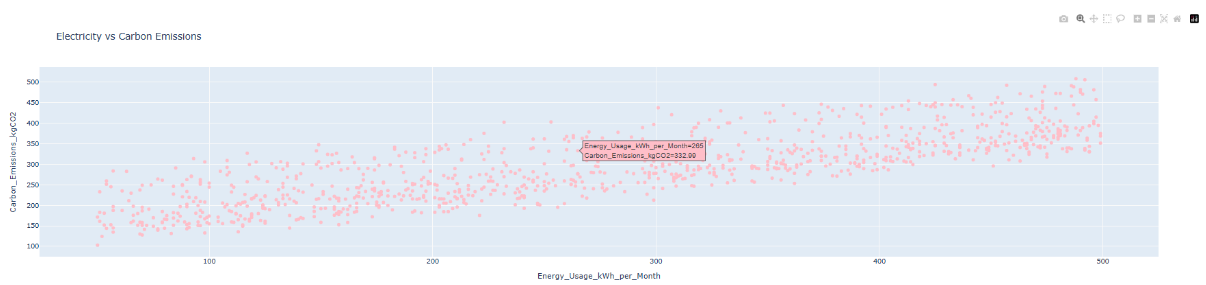
SDG Impact Analysis



Purpose:

- Demonstrates contribution to SDGs.
- Highlights sustainability impact.

Electricity Usage vs Carbon Emissions



Purpose:

- Shows the relationship between electricity usage and carbon emissions.
- Interactive visualization using Plotly.

TECHNOLOGIES USED

Programming Language

- Python

Data Science

- Pandas
- NumPy

Machine Learning

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Data Visualization

- Matplotlib
- Seaborn
- Plotly

Development Tools

- Google Colab
- Jupyter Notebook
- VS Code

Front-End Technologies

- HTML
- CSS
- JavaScript

DESIGN THINKING APPROACH

Empathize

Understanding sustainability challenges faced by students.

Define

Identifying the need for carbon footprint awareness.

Ideate

Developing an AI-powered sustainability solution.

Prototype

Creating Machine Learning models and dashboards.

Test

Evaluating model performance and user experience.

RESPONSIBLE AI PRINCIPLES

Fairness

The system evaluates all users using the same prediction methodology.

Transparency

Prediction factors are clearly explained to users.

Privacy

No personal or sensitive information is collected.

Accountability

Predictions are provided only for educational and awareness purposes.

Ethical AI

The project promotes environmental sustainability and social responsibility.

RESULTS

The developed AI system successfully:

- Predicted student carbon emissions
- Identified major emission sources
- Generated sustainability recommendations
- Improved environmental awareness
- Supported SDG 13 – Climate Action
- Demonstrated practical implementation of AI for Sustainability

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IMPACT ANALYSIS

Environmental Impact

- Increased awareness of carbon emissions
- Encouraged sustainable behavior
- Reduced environmental impact
- Promoted climate action

Social Impact

- Enhanced sustainability education
- Increased student participation
- Improved environmental responsibility

Economic Impact

- Reduced resource consumption
- Improved energy efficiency
- Better utilization of resources

INNOVATION OF THE PROJECT

Unlike traditional awareness programs, this project combines:

- Artificial Intelligence
- Machine Learning
- Sustainability Analytics
- Interactive Dashboards
- Environmental Education

This enables personalized carbon footprint estimation and actionable sustainability recommendations.

FUTURE ENHANCEMENTS

- Mobile Application Development
- Real-Time Carbon Monitoring
- AI Sustainability Chatbot
- Campus Sustainability Dashboard
- Personalized Eco Challenges
- IoT-Based Smart Monitoring
- Cloud Deployment
- Multi-Language Support

CONCLUSION

The Carbon Footprint Tracker for Student Lifestyle Using AI demonstrates how Artificial Intelligence can be used responsibly to address real-world sustainability challenges.

By combining Machine Learning, sustainability analytics, and interactive visualization, the project empowers students to understand their environmental impact and adopt sustainable lifestyle practices.

The project strongly supports SDG 13 – Climate Action and showcases the potential of AI in creating meaningful environmental and social impact.

IMPACT STATEMENT

If implemented across educational institutions, this solution can help thousands of students understand their environmental impact, reduce carbon emissions, adopt sustainable lifestyle practices, and contribute toward global climate action goals.

The project demonstrates how Artificial Intelligence can be used responsibly to create measurable environmental, social, and educational impact.

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8. Plotly Documentation
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10. Sustainability and Climate Action Research Articles

DEVELOPED BY

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